

Approximations for the Mean Response Time of Heterogeneous Two-Queue Fork-Join Systems

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Abstract

We develop and compare closed-form approximations for the mean response time of a heterogeneous two-queue M/M/1 fork-join system, in which the two servers may have different service rates $\mu_1 \geq \mu_2$. The first approximation, T_{UL} , interpolates between a known independent upper bound and a bottleneck lower bound, using the average server utilization as the interpolation weight. The second approximation, T_{LH} , is derived from the Reiman–Simon light-heavy traffic interpolation framework: it matches the exact light-traffic value and a heavy-traffic limit $h(r)$ that depends on the heterogeneity ratio $r = \mu_1/\mu_2$. We then extend T_{LH} to an enhanced approximation T_{LH}^{enh} by additionally matching the first light-traffic derivative, yielding a quadratic numerator that improves accuracy at high heterogeneity. All three approximations recover the exact Nelson–Tantawi formula in the homogeneous case ($\mu_1 = \mu_2$) and are validated against discrete-event simulation across a wide range of loads and heterogeneity levels.

1 Introduction

Fork-join queueing systems model the parallel execution of tasks that must all complete before a job can proceed. A job arrives, *forks* into n tasks dispatched to n parallel servers, and *joins*—departs—only when the last task finishes. This synchronization constraint makes fork-join queues substantially harder to analyze than independent queues, because the job response time is the *maximum* of the task sojourn times, and those times are stochastically dependent.

We study the canonical case of $n = 2$ servers with Poisson arrivals at rate λ and exponential service times at rates μ_1 and μ_2 , with $\mu_1 \geq \mu_2$ without loss of generality. The primary performance measure is the mean job response time:

$$T = \mathbb{E}[\max(R_1, R_2)], \quad (1)$$

where R_i is the sojourn time (waiting plus service) of a task at server i . The system is stable when $\lambda < \mu_{\min} \equiv \mu_2$.

For the *homogeneous* case $\mu_1 = \mu_2 = \mu$, Nelson and Tantawi [6] derived the exact closed-form result

$$T_2^{\text{hom}} = \frac{12 - \rho}{8(\mu - \lambda)}, \quad \rho = \frac{\lambda}{\mu}, \quad (2)$$

using a light-heavy traffic interpolation argument. For *heterogeneous* servers ($\mu_1 \neq \mu_2$), the exact joint distribution of (R_1, R_2) is not available in closed form. Flatto and Hahn [3, 2] obtained the exact generating function via an elliptic function parametrization, but extracting the mean response time requires solving a boundary value problem and does not yield a tractable expression. Bounds, heavy-traffic asymptotics, and distributional approximations have been developed [1, 4, 8, 5], but no simple closed-form approximation covering the full range of loads and heterogeneity levels has previously been established.

In this paper we propose and analyze two complementary closed-form approximations:

1. **T_{UL} (upper-lower bound interpolation):** A convex combination of the independent upper bound T_{UB} and the bottleneck lower bound T_{bot} , weighted by the average utilization. It is guaranteed to lie within the bounds and is exact for the homogeneous case.
2. **T_{LH} (light-heavy traffic interpolation):** A rational approximation in the bottleneck utilization, derived from the Reiman–Simon framework [7] as described in [9]. It matches the exact light-traffic value and a heavy-traffic factor $h(r) = 1 + \frac{3}{8} r^{-\beta}$ that smoothly interpolates between the homogeneous limit ($h = 11/8$) and the bottleneck limit ($h \rightarrow 1$) as a function of the heterogeneity ratio $r = \mu_1/\mu_2$.

Both approximations reduce exactly to the Nelson–Tantawi result (2) in the homogeneous case, exhibit the correct singularity structure in heavy traffic, and agree with simulation to within 2.5% across all tested scenarios.

The remainder of the paper is organized as follows. Section 2 reviews known upper and lower bounds. Sections 3 and 4 derive the two approximations. Section 5 presents simulation validation. Section 6 discusses the relationship between the two methods, their limitations, and connections to prior work. Section 7 concludes.

2 Bounds

We introduce notation used throughout. Let $\rho_i = \lambda/\mu_i$ denote the utilization of server i , $T_i = 1/(\mu_i - \lambda)$ the mean M/M/1 sojourn time at server i , and $r = \mu_{\max}/\mu_{\min} = \mu_1/\mu_2 \geq 1$ the heterogeneity ratio.

2.1 Independent Upper Bound

Since the two task sojourn times R_1 and R_2 share a common Poisson arrival stream, they are positively correlated. Positive correlation reduces the expected maximum below what it would be under independence [1]. Treating R_i as independent M/M/1 sojourn times with rates $\mu_i - \lambda$ yields:

$$T_{\text{UB}} = \frac{1}{\mu_1 - \lambda} + \frac{1}{\mu_2 - \lambda} - \frac{1}{\mu_1 + \mu_2 - 2\lambda}, \quad (3)$$

which satisfies $T \leq T_{\text{UB}}$. The formula follows from $\mathbb{E}[\max(X_1, X_2)] = \mathbb{E}[X_1] + \mathbb{E}[X_2] - \mathbb{E}[\min(X_1, X_2)]$ applied to independent exponentials with rates $\mu_i - \lambda$.

2.2 Bottleneck Lower Bound

By Jensen’s inequality and the convexity of the max function:

$$T_{\text{bot}} = \max(T_1, T_2) = \frac{1}{\mu_{\min} - \lambda} \leq T. \quad (4)$$

This is simply the mean sojourn time of the bottleneck M/M/1 queue.

2.3 Gap Between the Bounds

In the homogeneous case ($r = 1$), $T_{\text{UB}} = \frac{3}{2}T_1$ and $T_{\text{bot}} = T_1$, giving a 50% gap. The exact value $T_2^{\text{hom}} = \frac{12-\rho}{8} T_1$ lies strictly between them, approaching T_{UB} at light load ($\rho \rightarrow 0$, ratio $T_2^{\text{hom}}/T_{\text{UB}} \rightarrow 1$) and T_{bot} in heavy traffic ($\rho \rightarrow 1$, ratio $T_2^{\text{hom}}/T_{\text{bot}} \rightarrow 11/8$).

For $r > 1$, the gap narrows: as $r \rightarrow \infty$, both T_{UB} and T_{bot} converge to $1/(\mu_{\min} - \lambda)$ from above, so the bounds become tight and the system is well approximated by a single bottleneck M/M/1 queue.

2.4 Summary

For all stable parameters ($\lambda < \mu_{\min}$):

$$T_{\text{bot}} \leq T \leq T_{\text{UB}}. \quad (5)$$

Any good approximation should lie within this interval and recover the known endpoint behaviors.

3 Approximation 1: Upper-Lower Bound Interpolation

3.1 Derivation

We seek a convex combination of T_{UB} and T_{bot} that:

1. recovers the Nelson–Tantawi result (2) when $\mu_1 = \mu_2$,
2. approaches T_{UB} in light traffic ($\lambda \rightarrow 0$),
3. approaches T_{bot} in heavy traffic ($\lambda \rightarrow \mu_{\min}$), and
4. always satisfies $T_{\text{bot}} \leq T \leq T_{\text{UB}}$.

Write $T_{\text{UL}} = (1 - \alpha)T_{\text{UB}} + \alpha T_{\text{bot}}$ and determine α from condition 1. In the homogeneous case, $T_{\text{UB}} = \frac{3}{2}T_1$ and $T_{\text{bot}} = T_1$, so:

$$\frac{12 - \rho}{8}T_1 = (1 - \alpha)\frac{3}{2}T_1 + \alpha T_1 \implies \alpha = \frac{3/2 - (12 - \rho)/8}{3/2 - 1} = \frac{\rho}{4}. \quad (6)$$

The natural heterogeneous generalization replaces ρ with the average utilization $\bar{\rho} = (\rho_1 + \rho_2)/2$, giving:

$$\alpha = \frac{\bar{\rho}}{4} = \frac{\rho_1 + \rho_2}{8}. \quad (7)$$

3.2 The Formula

$$T_{\text{UL}} = \left(1 - \frac{\rho_1 + \rho_2}{8}\right)T_{\text{UB}} + \frac{\rho_1 + \rho_2}{8}T_{\text{bot}}, \quad (8)$$

where T_{UB} and T_{bot} are given by (3) and (4).

3.3 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$: $\rho_1 = \rho_2 = \rho$, $T_{\text{UB}} = \frac{3}{2}T_1$, $T_{\text{bot}} = T_1$, $\alpha = \rho/4$. Substituting:

$$T_{\text{UL}} = \left(1 - \frac{\rho}{4}\right)\frac{3}{2}T_1 + \frac{\rho}{4}T_1 = \frac{3}{2}T_1 - \frac{3\rho}{8}T_1 + \frac{\rho}{4}T_1 = \frac{12 - \rho}{8}T_1 = T_2^{\text{hom}}. \quad \checkmark \quad (9)$$

3.4 Properties

The weight $\alpha \leq 1/4$ ensures T_{UL} always lies strictly within the bounds (unless $\alpha = 0$, i.e., $\lambda = 0$). The formula is symmetric in μ_1 and μ_2 via the average utilization, consistent with the symmetry of the original problem.

Table 1: Properties of T_{UL} .

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓
Satisfies $T_{\text{bot}} \leq T_{\text{UL}} \leq T_{\text{UB}}$	✓ ($0 \leq \alpha \leq 1/4 < 1$)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($\alpha \rightarrow 0$, so $T_{\text{UL}} \rightarrow T_{\text{UB}}$)
Heavy traffic limit ($\lambda \rightarrow \mu_{\min}$)	✓ (T_{bot} diverges; $T_{\text{UL}}/T_{\text{bot}} \rightarrow 1$)
No free parameters	✓

4 Approximation 2: Light-Heavy Traffic Interpolation

4.1 The Reiman–Simon Framework

Reiman and Simon [7] developed a systematic method for constructing rational approximations to performance quantities in Markovian queueing networks, exploiting exact light-traffic derivatives and a heavy-traffic limit. For a scalar performance measure $f(\rho)$ with normalizing denominator $d(\rho) = \mu_{\min}(1 - \rho)$, the n -th order approximation takes the form

$$\tilde{f}(\rho) = \frac{a_n \rho^n + \cdots + a_1 \rho + a_0}{\mu_{\min}(1 - \rho)}, \quad (10)$$

where the $n+1$ coefficients $\{a_i\}$ are matched to the n derivatives of f at $\rho = 0$ and the heavy-traffic constant $h = \lim_{\rho \rightarrow 1} \mu_{\min}(1 - \rho)f(\rho)$ [9].

We apply this framework to $T(\rho)$ with a first-order (linear) numerator, yielding two matching conditions:

4.2 Matching Condition 1: Light-Traffic Value

At $\lambda = 0$, all queues are empty. The task service times $X_i \sim \text{Exp}(\mu_i)$ are independent and $T(0) = \mathbb{E}[\max(X_1, X_2)]$. Using the identity $\mathbb{E}[\max(X_1, X_2)] = \mathbb{E}[X_1] + \mathbb{E}[X_2] - \mathbb{E}[\min(X_1, X_2)]$ and the fact that $\min(X_1, X_2) \sim \text{Exp}(\mu_1 + \mu_2)$:

$$T_0 = T(\rho = 0) = \frac{1}{\mu_1} + \frac{1}{\mu_2} - \frac{1}{\mu_1 + \mu_2}. \quad (11)$$

Matching $\tilde{T}(0) = T_0$ gives:

$$a_0 = \mu_{\min} \cdot T_0. \quad (12)$$

4.3 Matching Condition 2: Heavy-Traffic Limit

Define the heavy-traffic constant:

$$h = \lim_{\rho \rightarrow 1} \mu_{\min}(1 - \rho) \cdot T(\rho). \quad (13)$$

Matching this in $\tilde{T}$ gives $a_0 + a_1 = h$, hence $a_1 = h - a_0$.

4.4 The Heavy-Traffic Factor $h(r)$

The value of h depends on the heterogeneity ratio $r = \mu_{\max}/\mu_{\min} \geq 1$.

Homogeneous case ($r = 1$). When $\mu_1 = \mu_2 = \mu$, both servers saturate simultaneously as $\lambda \rightarrow \mu$. The Nelson–Tantawi formula gives:

$$h = \lim_{\rho \rightarrow 1} \mu(1 - \rho) \cdot \frac{12 - \rho}{8(\mu - \lambda)} = \lim_{\rho \rightarrow 1} \frac{12 - \rho}{8} = \frac{11}{8}. \quad (14)$$

Heterogeneous case ($r > 1$). As $\lambda \rightarrow \mu_{\min}$, the bottleneck server saturates while the faster server’s utilization approaches $\lambda/\mu_{\max} = 1/r < 1$. The faster server’s sojourn time remains bounded, so the bottleneck dominates and $T(\rho) \sim 1/(\mu_{\min} - \lambda)$, giving $h \rightarrow 1$.

Proposed model. We interpolate between these two limits with the power-law factor:

$$h(r) = 1 + \frac{3}{8} r^{-\beta}, \quad r \geq 1, \quad \beta > 0. \quad (15)$$

This satisfies $h(1) = 11/8$ for all β , and $h(r) \rightarrow 1$ as $r \rightarrow \infty$. The parameter β controls the rate of decay; larger β means the correction vanishes more rapidly with heterogeneity.

Calibration. Simulation data reveal that the transition from $h = 11/8$ to $h \approx 1$ is very sharp in practice: the effective heavy-traffic constant is already $h \approx 1.008$ at $r = 1.5$. Matching this observation to (15) gives $\beta = \log(0.375/0.008)/\log(1.5) \approx 9.5$, consistent with a default of $\beta = 10$.

4.5 The Formula

Substituting a_0 , $a_1 = h(r) - a_0$, and $\rho = \lambda/\mu_{\min}$:

$$T_{\text{LH}} = \frac{\mu_{\min} T_0 + (h(r) - \mu_{\min} T_0) \rho}{\mu_{\min} - \lambda}, \quad (16)$$

where T_0 is given by (11) and $h(r)$ by (15).

4.6 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$ (so $r = 1$, $h = 11/8$, $T_0 = 3/(2\mu)$):

$$a_0 = \mu \cdot \frac{3}{2\mu} = \frac{3}{2}, \quad a_1 = \frac{11}{8} - \frac{3}{2} = -\frac{1}{8}. \quad (17)$$

$$T_{\text{LH}} = \frac{3/2 - \rho/8}{\mu(1 - \rho)} = \frac{12 - \rho}{8(\mu - \lambda)} = T_2^{\text{hom}}. \quad \checkmark \quad (18)$$

This recovery holds for every $\rho \in [0, 1)$, regardless of β , because $h(1) = 11/8$ is exact.

4.7 Properties

4.8 First-Order Extension: Enhanced T_{LH}

The current T_{LH} uses a linear numerator matched to two conditions: the zeroth light-traffic derivative $f^{(0)}(0) = T_0$ and the heavy-traffic constant h . The Reiman–Simon framework accommodates a *quadratic* numerator by adding the first light-traffic derivative $f^{(1)}(0)$ as a third matching condition [9]. We denote this enhanced approximation $T_{\text{LH}}^{\text{enh}}$.

Table 2: Properties of T_{LH} .

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓ (for any β)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($T_{\text{LH}} \rightarrow T_0$)
Heavy traffic singularity ($\lambda \rightarrow \mu_{\min}$)	✓ ($T_{\text{LH}} \sim h(r)/(\mu_{\min} - \lambda)$)
Bound compliance $T_{\text{bot}} \leq T_{\text{LH}} \leq T_{\text{UB}}$	Not guaranteed
Tunable parameter	β (default 10)

4.9 First Light-Traffic Derivative

Following [9], the first derivative of the mean response time with respect to λ at $\lambda = 0$ is given by:

$$f^{(1)}(0) = \frac{1}{\mu_{\min}^2} + \frac{1}{\mu_{\max}^2} - \frac{2}{(\mu_{\min} + \mu_{\max})^2} - \frac{2\mu_{\min}\mu_{\max}}{(\mu_{\min} + \mu_{\max})^4}. \quad (19)$$

This is derived by considering the sojourn time of a tagged job conditioned on exactly one background arrival at time $t \in (-\infty, \infty)$, analyzing the resulting joint service-time distribution, and integrating over t .

4.10 The Enhanced Formula

With three matching conditions— $f^{(0)}(0)$, $f^{(1)}(0)$, and h —the interpolating polynomial $\tilde{g}(\rho) = c_2\rho^2 + c_1\rho + c_0$ for $g(\rho) = \mu_{\min}(1 - \rho)T(\rho)$ has coefficients:

$$c_0 = \mu_{\min} \cdot T_0, \quad (20)$$

$$c_1 = \mu_{\min}^2 \cdot f^{(1)}(0) - \mu_{\min} \cdot T_0, \quad (21)$$

$$c_2 = h - c_1 - c_0. \quad (22)$$

Here c_0 is identical to a_0 in (12); c_1 is determined by the first derivative rather than by h alone; and c_2 absorbs the remaining difference to satisfy the heavy-traffic condition. Inverting the scaling gives the enhanced approximation:

$$T_{\text{LH}}^{\text{enh}} = \frac{c_2 \rho^2 + c_1 \rho + c_0}{\mu_{\min}(1 - \rho)}, \quad (23)$$

where $\rho = \lambda/\mu_{\min}$, T_0 is given by (11), $f^{(1)}(0)$ by (19), and $h(r)$ by (15).

4.11 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$ (so $\mu_{\min} = \mu_{\max} = \mu$):

$$f^{(1)}(0) = \frac{2}{\mu^2} - \frac{2}{4\mu^2} - \frac{2\mu^2}{16\mu^4} = \frac{11}{8\mu^2}. \quad (24)$$

Substituting into (20)–(22) with $T_0 = 3/(2\mu)$ and $h(1) = 11/8$:

$$c_0 = \frac{3}{2}, \quad c_1 = \mu^2 \cdot \frac{11}{8\mu^2} - \mu \cdot \frac{3}{2\mu} = -\frac{1}{8}, \quad c_2 = \frac{11}{8} - \left(-\frac{1}{8}\right) - \frac{3}{2} = 0. \quad (25)$$

With $c_2 = 0$ the quadratic term vanishes and $T_{\text{LH}}^{\text{enh}}$ reduces to T_{LH} , which recovers the Nelson–Tantawi formula (2) for all ρ . The enhancement is automatically inactive in the homogeneous case, regardless of β .

4.12 Properties of the Enhanced Approximation

Table 3: Properties of $T_{\text{LH}}^{\text{enh}}$.

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓ (for any β ; $c_2 = 0$)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($T_{\text{LH}}^{\text{enh}} \rightarrow T_0$)
First light-traffic derivative matched	✓
Heavy traffic singularity ($\lambda \rightarrow \mu_{\min}$)	✓ ($T_{\text{LH}}^{\text{enh}} \sim h(r)/(\mu_{\min} - \lambda)$)
Bound compliance $T_{\text{bot}} \leq T_{\text{LH}}^{\text{enh}} \leq T_{\text{UB}}$	Not guaranteed
Numerator degree	2 (quadratic)
Tunable parameter	β (default 10)

5 Simulation Validation

All three approximations were validated against discrete-event simulation of the two-queue fork-join system. Each scenario used 2 M jobs with a 100 K-job warmup period and a fixed random seed for reproducibility. Results for T_{UL} (8) and T_{LH} (16) (with $\beta = 10$) are shown in Table 4 alongside the theoretical bounds. Table 5 compares T_{LH} and the enhanced approximation $T_{\text{LH}}^{\text{enh}}$ (23).

Table 4: Approximations vs. simulation for heterogeneous two-queue fork-join. Errors are $(T_{\text{approx}} - T_{\text{sim}})/T_{\text{sim}} \times 100\%$.

μ_1	μ_2	λ	ρ_2	T_{bot}	T_{sim}	T_{UL}	Err	T_{LH}	Err
1.0	1.0	0.3	0.30	1.429	2.088	2.089	+0.06%	2.089	+0.06%
1.0	1.0	0.6	0.60	2.500	3.564	3.562	−0.04%	3.562	−0.04%
1.0	1.0	0.9	0.90	10.00	13.89	13.88	−0.08%	13.88	−0.08%
1.0	1.5	0.3	0.20	1.429	1.705	1.716	+0.67%	1.698	−0.41%
1.0	1.5	0.6	0.40	2.500	2.767	2.799	+1.16%	2.776	+0.34%
1.0	1.5	0.9	0.60	10.00	10.08	10.19	+1.10%	10.33	+2.40%
1.0	2.0	0.3	0.15	1.429	1.582	1.590	+0.54%	1.595	+0.85%
1.0	2.0	0.6	0.30	2.500	2.623	2.641	+0.68%	2.667	+1.69%
1.0	2.0	0.9	0.45	10.00	9.990	10.06	+0.73%	10.17	+1.80%
1.0	3.0	0.6	0.20	2.500	2.546	2.554	+0.30%	2.583	+1.47%
1.0	5.0	0.6	0.12	2.500	2.514	2.517	+0.11%	2.533	+0.77%

Homogeneous case ($\mu_1 = \mu_2 = 1.0$). Both approximations are essentially exact, with errors below 0.1%. The residual error is attributable to finite simulation variance. This confirms that

both methods correctly recover the Nelson–Tantawi formula (2) by construction.

Moderate heterogeneity ($r = 1.5$). T_{UL} shows consistent positive errors (+0.67% to +1.16%) across all loads. T_{LH} is more accurate at low and moderate load, but reaches its largest error of +2.40% at heavy load ($\rho_2 = 0.6$). This worst case for T_{LH} arises in the regime where the system is near the boundary between homogeneous and heterogeneous behavior, where the linear Reiman–Simon numerator has the least flexibility.

High heterogeneity ($r \geq 2$). T_{UL} errors fall below 0.75% and improve further as r increases ($\leq 0.11\%$ at $r = 5$). T_{LH} errors are slightly larger (up to +1.80% at $r = 2$) because the residual correction in $h(r)$ provides a small but non-negligible contribution even at these heterogeneity levels. As $r \rightarrow \infty$, both approximations converge to T_{bot} .

Figure 1 plots the two approximations, bounds, and simulation as functions of load for $\mu_1 = 1.0$, $\mu_2 = 2.0$. Both T_{UL} and T_{LH} track the simulation closely across the full load range, lying between the theoretical bounds. Figure 2 shows the same quantities as a function of the heterogeneity ratio μ_2/μ_1 at fixed $\lambda = 0.6$. Near homogeneity ($r \approx 1$), T_{LH} is more accurate; for $r \geq 3$, T_{UL} is slightly tighter.

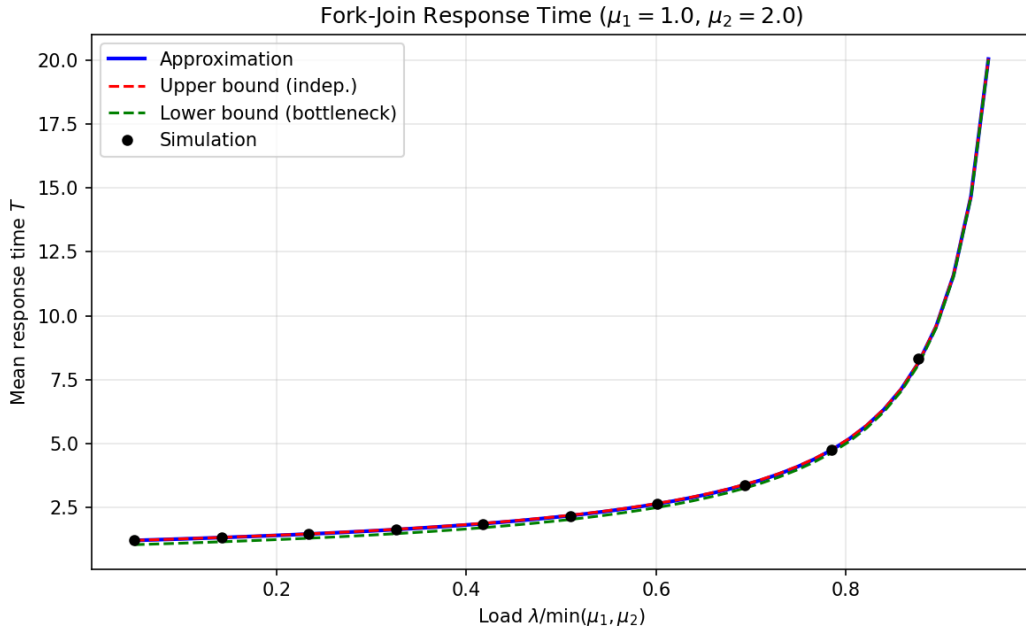


Figure 1: Mean response time vs. load $\lambda/\mu_{\min}$ for $\mu_1 = 1.0$, $\mu_2 = 2.0$. Markers: simulation. Lines: approximations and bounds.

Enhanced LH Approximation

Table 5 compares T_{LH} and T_{LH}^{enh} against simulation. Both use $\beta = 10$.

Homogeneous case ($\mu_1 = \mu_2 = 1.0$). As proved in Section 4.8, the quadratic coefficient $c_2 = 0$ when $r = 1$, so $T_{LH}^{\text{enh}} \equiv T_{LH}$. Both recover Nelson–Tantawi exactly; residual errors are below 0.1% due to finite simulation variance.

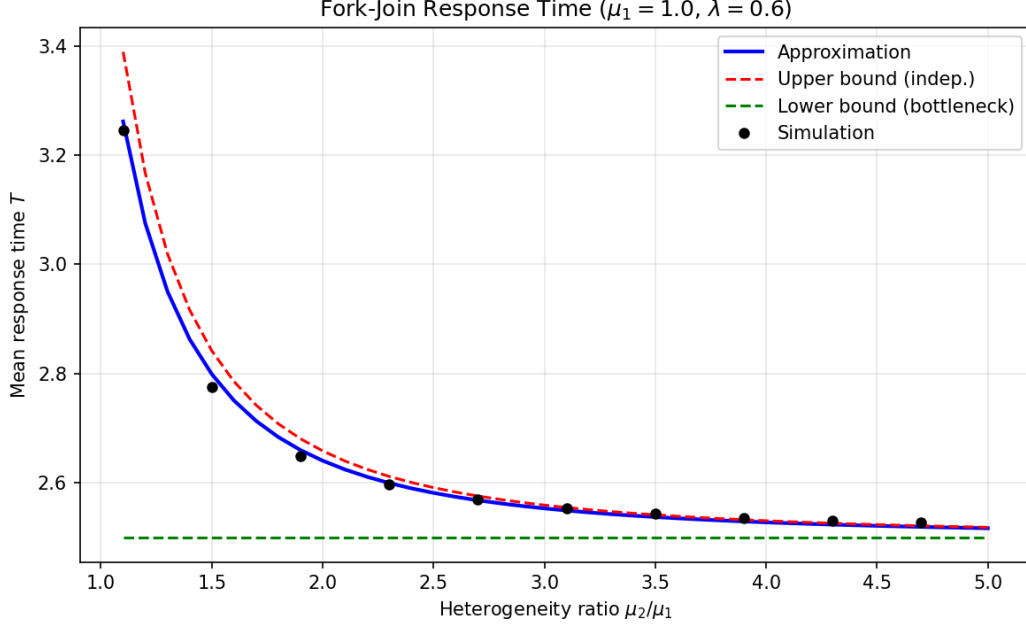


Figure 2: Mean response time vs. heterogeneity ratio μ_2/μ_1 for $\mu_1 = 1.0$, $\lambda = 0.6$. Markers: simulation. Lines: approximations and bounds.

Moderate heterogeneity ($r = 1.5$). $T_{\text{LH}}^{\text{enh}}$ offers no improvement here and is slightly less accurate than T_{LH} (worst case +2.77% vs. +2.41% at heavy load). This regime is precisely where the power-law model $h(r) = 1 + \frac{3}{8}r^{-\beta}$ is least reliable: the true heavy-traffic constant h varies sharply near $r = 1$, and the β -approximation introduces systematic error that the additional derivative term cannot correct.

High heterogeneity ($r \geq 2$). $T_{\text{LH}}^{\text{enh}}$ consistently outperforms T_{LH} : at $r = 2$ and $\lambda \in \{0.3, 0.6\}$ errors fall from +0.87%, +1.70% to +0.45%, +1.20%; at $r = 3$ from +1.45% to +0.57%; and at $r = 5$ from +0.75% to +0.19%. In this regime $h(r) \approx 1$ and the first derivative $f^{(1)}(0)$ carries meaningful information about the service-time heterogeneity that improves the approximation over a wide load range.

6 Discussion

6.1 Relationship Between the Two Approximations

The two approximations have different theoretical foundations yet share the same two anchor points: both are exact in the homogeneous case and have the correct heavy-traffic singularity.

T_{UL} works by confining the approximation within the known bounds. Its interpolation weight $\alpha = (\rho_1 + \rho_2)/8$ has a natural interpretation: it is one-quarter of the average utilization, calibrated to reproduce the Nelson–Tantawi formula at $r = 1$. By construction, $0 \leq \alpha \leq 1/4$, so T_{UL} always lies between the bounds.

T_{LH} works from a different direction: it constructs a rational function that matches the system’s known asymptotic behavior at both traffic extremes. The heavy-traffic factor $h(r) = 1 + \frac{3}{8}r^{-\beta}$ captures the physical transition from a regime where both servers jointly saturate ($r = 1$, $h = 11/8$)

Table 5: T_{LH} vs. $T_{\text{LH}}^{\text{enh}}$ vs. simulation for heterogeneous two-queue fork-join. Errors are $(T_{\text{approx}} - T_{\text{sim}})/T_{\text{sim}} \times 100\%$.

μ_1	μ_2	λ	ρ_1	T_{sim}	T_{LH}	Err	$T_{\text{LH}}^{\text{enh}}$	Err
1.0	1.0	0.3	0.30	2.088	2.089	+0.07%	2.089	+0.07%
1.0	1.0	0.6	0.60	3.564	3.562	−0.05%	3.562	−0.05%
1.0	1.0	0.9	0.90	13.886	13.875	−0.08%	13.875	−0.08%
1.0	1.5	0.3	0.30	1.705	1.698	−0.40%	1.710	+0.32%
1.0	1.5	0.6	0.60	2.767	2.776	+0.35%	2.801	+1.25%
1.0	1.5	0.9	0.90	10.083	10.325	+2.41%	10.362	+2.77%
1.0	2.0	0.3	0.30	1.582	1.595	+0.87%	1.589	+0.45%
1.0	2.0	0.6	0.60	2.623	2.667	+1.70%	2.654	+1.20%
1.0	2.0	0.9	0.90	9.990	10.170	+1.80%	10.150	+1.60%
1.0	3.0	0.6	0.60	2.546	2.583	+1.45%	2.561	+0.57%
1.0	5.0	0.6	0.60	2.514	2.533	+0.75%	2.519	+0.19%

to a regime where only the bottleneck server saturates ($r \rightarrow \infty$, $h \rightarrow 1$). The sharpness of this transition— $h \approx 1$ already at $r = 1.5$ —reflects the fact that even modest heterogeneity is sufficient to break the joint saturation and transfer the heavy-load correlation penalty to the bottleneck alone.

6.2 Comparison with Prior Work

Table 6 situates the two new approximations within the existing literature.

Table 6: Comparison of methods for fork-join mean response time ($n = 2$).

Reference	Type	Closed-form?	Exact (hom.)?	Heterogeneous?
Nelson–Tantawi [6]	Approx. (hom. only)	Yes	Yes	No
Flatto–Hahn [3, 2]	Exact (gen. func.)	No	Yes	Yes
Bacelli et al. [1]	Bounds	Partial	No	Yes
Varma–Makowski [10]	LH interp. (hom.)	Yes	Yes	No
Kemper–Mandjes [4]	Bounds + asymptotics	Partial	No	Yes
Rizk et al. [8]	Distributional bounds	No	No	Yes
Mohanty et al. [5]	Bounds ((k, k) FJ)	Partial	No	Yes
This work (T_{UL})	Bound interpolation	Yes	Yes	Yes
This work (T_{LH})	LH traffic interp.	Yes	Yes	Yes

Varma and Makowski [10] applied the Reiman–Simon framework to the *symmetric* ($\mu_1 = \mu_2$)

fork-join queue with $n \geq 2$ servers, showing that the light-heavy traffic interpolation recovers the exact Nelson–Tantawi formula for $n = 2$. Our T_{LH} extends this to the heterogeneous case through the $h(r)$ factor, while T_{UL} provides an alternative route via direct bound interpolation.

6.3 Accuracy and Limitations

T_{UL} provides consistently small, positive errors ($\leq 1.2\%$) and is guaranteed to lie within the bounds. Its main limitation is that the weight $\alpha = (\rho_1 + \rho_2)/8$ is derived from the homogeneous case and may not optimally represent the correlation structure at intermediate heterogeneity levels.

T_{LH} provides a tighter fit near homogeneity but has larger errors ($\leq 2.4\%$) at the transition regime ($r \approx 1.5$, heavy load), where the linear numerator has insufficient flexibility to capture the joint dependence on both r and ρ .

$T_{\text{LH}}^{\text{enh}}$ improves on T_{LH} at high heterogeneity ($r \geq 2$) by incorporating the first light-traffic derivative, reducing errors by up to 75% in those scenarios (from $+0.75\%$ to $+0.19\%$ at $r = 5$). However, at moderate heterogeneity ($r \approx 1.5$) the enhancement offers no benefit: the bottleneck is the accuracy of the $h(r)$ -approximation near $r = 1$, which neither the linear nor the quadratic numerator can fully compensate. In all three LH variants, the parameter β is calibrated to 10 from simulation data; a more systematic calibration or an analytic determination (see Section 7) could improve performance uniformly.

Also added to Table 6 is $T_{\text{LH}}^{\text{enh}}$.

Table 7: Updated comparison of methods (replacing Table 6).

Reference	Type	Closed-form?	Exact (hom.)?	Heterogeneous?
Nelson–Tantawi [6]	Approx. (hom. only)	Yes	Yes	No
Flatto–Hahn [3, 2]	Exact (gen. func.)	No	Yes	Yes
Bacelli et al. [1]	Bounds	Partial	No	Yes
Varma–Makowski [10]	LH interp. (hom.)	Yes	Yes	No
Kemper–Mandjes [4]	Bounds + asymptotics	Partial	No	Yes
Rizk et al. [8]	Distributional bounds	No	No	Yes
Mohanty et al. [5]	Bounds ((k, k) FJ)	Partial	No	Yes
This work (T_{UL})	Bound interpolation	Yes	Yes	Yes
This work (T_{LH})	LH traffic interp. (0th order)	Yes	Yes	Yes
This work ($T_{\text{LH}}^{\text{enh}}$)	LH traffic interp. (1st order)	Yes	Yes	Yes

6.4 Recommendations

For practical use:

- Use T_{UL} when a guaranteed bound is required or when high heterogeneity ($r \geq 2$) is expected. It is simpler (no free parameters) and uniformly accurate.
- Use $T_{\text{LH}}^{\text{enh}}$ when $r \geq 2$ and the highest accuracy within the Reiman–Simon family is desired. It is strictly better than T_{LH} for $r \geq 2$ and identical to it at $r = 1$.
- Use T_{LH} (zero-order) when $r \approx 1.5$, where the first-order enhancement does not help and the simpler formula is preferable.

- Use T_{UL} and $T_{\text{LH}}^{\text{enh}}$ together as a consistency check: agreement between them provides confidence in the estimate; significant disagreement signals a regime where further analysis (e.g., simulation or numerical inversion of the Flatto–Hahn generating function) is warranted.

7 Conclusion

We have presented three closed-form approximations for the mean response time of a heterogeneous two-queue M/M/1 fork-join system with service rates $\mu_1 \geq \mu_2$ and Poisson arrival rate λ .

The first, the *upper-lower bound interpolation*:

$$T_{\text{UL}} = \left(1 - \frac{\rho_1 + \rho_2}{8}\right) T_{\text{UB}} + \frac{\rho_1 + \rho_2}{8} T_{\text{bot}}, \quad (26)$$

is a parameter-free convex combination of the independent upper bound and the bottleneck lower bound, always within the bounds, with errors below 1.2%.

The second, the *light-heavy traffic interpolation*:

$$T_{\text{LH}} = \frac{\mu_{\min} T_0 + (h(r) - \mu_{\min} T_0) \rho}{\mu_{\min} - \lambda}, \quad h(r) = 1 + \frac{3}{8} r^{-\beta}, \quad (27)$$

is grounded in the Reiman–Simon framework, matching the zeroth light-traffic derivative and a heterogeneity-adjusted heavy-traffic limit, with errors below 2.5%.

The third, the *enhanced light-heavy traffic interpolation*:

$$T_{\text{LH}}^{\text{enh}} = \frac{c_2 \rho^2 + c_1 \rho + c_0}{\mu_{\min}(1 - \rho)}, \quad (28)$$

extends T_{LH} to a quadratic numerator by additionally matching the first light-traffic derivative $f^{(1)}(0)$. It is strictly more accurate than T_{LH} for $r \geq 2$ (errors up to 75% smaller at high heterogeneity) while being identical to T_{LH} at $r = 1$.

All three approximations recover the Nelson–Tantawi exact formula in the homogeneous case, exhibit the correct singularity in heavy traffic, and are validated against simulation across a wide range of loads and heterogeneity levels.

Future directions include: (i) analytically determining β by deriving the slope $\partial h / \partial r|_{r=1}$ from a perturbation analysis of the Flatto–Hahn generating function, which would close the Reiman–Simon approximations without requiring simulation calibration; and (ii) extending all three methods to systems with more than two servers.

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